MILITARY SPECIFICATION

CLEANING MACHINE, SMALL AIRCRAFT, MOBILE

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers one type of mobile (tractor towed) aircraft cleaning machine for use in removing salt deposits, oil, grease and other contaminants from aircraft surfaces. (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 Government documents normally furnished - The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-766	Steel Plate, Sheet, and Strip-Corrosion Resisting
TT-C-490	Cleaning Method and Pretreatment of Ferrous Surfaces for Organic Coatings
TT-P-664	Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting
VV-K-211	Kerosene
WW-C-624	Coupling Assembly, Hose (Fire, Garden, Water Suction)
ZZ-H-611	Hose, Water Wrapped
PPP-B-601	Boxes, Wood, Cleated, Plywood

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PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-T-76	Tape, Pressure-Sensitive, Adhesive Paper, Water Resistant, (For Carton Sealing)

Military

MIL-P-116	Preservation, Methods of
MIL-V-173	Varnish, Moisture and Fungus Resistant (For the Treatment of Communication, Electronic, and Associated Electrical Equipment)
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-P-1144	Pipe, Stainless Steel, Corrosion Resistant, Seamless or Welded
MIL-F-3541	Fittings, Lubrication
MIL-E-7729	Enamel, Gloss
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For
MIL-M-7793	Meter, Time Totalizing
MIL-W-8005	Wheels and Hubs, for Industrial Pneumatic Tires
MIL-M-8090	Mobility Requirements, Ground Support Equipment, General Specification For
MIL-A-8421	Air Transportability Requirements, General Specification For
MIL-H-8794	Hose; Rubber, Hydraulic, Fuel and Oil Resistant

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leg with caster. The landing leg shall be designed to support the trailer in

MIL-H-8795	Hose Assemblies, Rubber, Hydraulic, Fuel and Oil Resistant
MIL-E-10062	Engines, Preparation for Shipment and Storage of
MIL-E-11275	Engines, Gasoline, Industrial Type, General Specification For
MIL-R-12144	Reflector, Indicating, Clearance
MIL-T-21863	Truck, Lift, Fork, Diesel, Pneumatic-Tire, All Purpose, 6000-Pound Capacity at 24-Inch Load Center
MIL-C-38334	Corrosion Removing Compound, Prepaint, For Aircraft Aluminum Surfaces
MIL-M-38784	Manuals, Technical, General Requirements For Preparation of
MIL-C-43616	Cleaning Compound, Aircraft Surface
MIL-T-83926	Tires, Pneumatic, Special Purpose (FOD)

STANDARDS

Federal

FED-STD-595	Colors
FED-STD-601	Rubber, Sampling and Testing

Military

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of

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MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS33586	Metals, Definition of Dissimilar
MS35000	Battery, Storage, Lead-Acid, Waterproof
MS51336	Lunette-Coupler, Drawbar, Ring

(When requesting applicable documents, refer to both title and number. Copies of unclassified documents may be obtained from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. Requests for copies of classified documents should be addressed to the Naval Publications and Forms Center, via the cognizant Government representative.)

DRAWINGS

Air Force

48B7796	Ring Assembly, Tie-Down 10,000 lb.

(Copies of Air Force drawings required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article - The cleaning machine furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein (see 4.3).

3.2 Selection of specifications and standards - Specifications and standards for necessary commodities and services herein shall be selected in accordance with MIL-STD-143.

3.3 Materials - Materials shall conform to the applicable specifications and shall be as specified herein and on the applicable drawings. Materials which are not covered by applicable specifications or standards shall be of the best quality, and entirely suitable for the purpose intended.

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3.3.1 Metals - Metals shall be corrosion-resistant or treated to resist corrosion caused by fuels, oils, salt spray, water, or atmospheric conditions. Magnesium alloy shall not be used in the construction of the cleaning machine.

3.3.1.1 Dissimilar metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.3.2 Fungus proof materials - Materials that are nutrients for fungi shall not be used.

3.3.3 Piping - The piping shall be in accordance with MIL-P-1144, composition 304.

3.3.4 Nonmetallic materials - Any nonmetallic material that is adversely affected by contact with cleaning compounds shall not be used.

3.4 Figures and manufacturer's part numbers - Figure 1, and the manufacturer's part numbers referenced herein are intended to be descriptive and not restrictive; and are not intended to preclude the use of any item otherwise conforming to this specification.

3.5 Design and construction - The aircraft cleaning machine shall be designed and constructed to provide a convenient and safe method of applying a water soluble cleaning compound and clear rinse water to the external surfaces of small aircraft. The design shall also include a foam system capable of applying the cleaning compound, as a foam that will cling to the vertical surfaces of aircraft.

3.5.1 Dimensions - The overall dimensions of the cleaning machine shall not exceed the following:

Length	- 90 inches (less towbar)
Width	- 55 inches
Height	- 42 inches

3.5.2 Weight - The weight of the cleaning machine (cleaning compound tank empty) shall not exceed 1100 pounds.

3.5.3 Power requirements - The cleaning machine shall be self-contained except for the water and high pressure air supply. The machine shall be powered by a gasoline engine. No external source of power shall be required.

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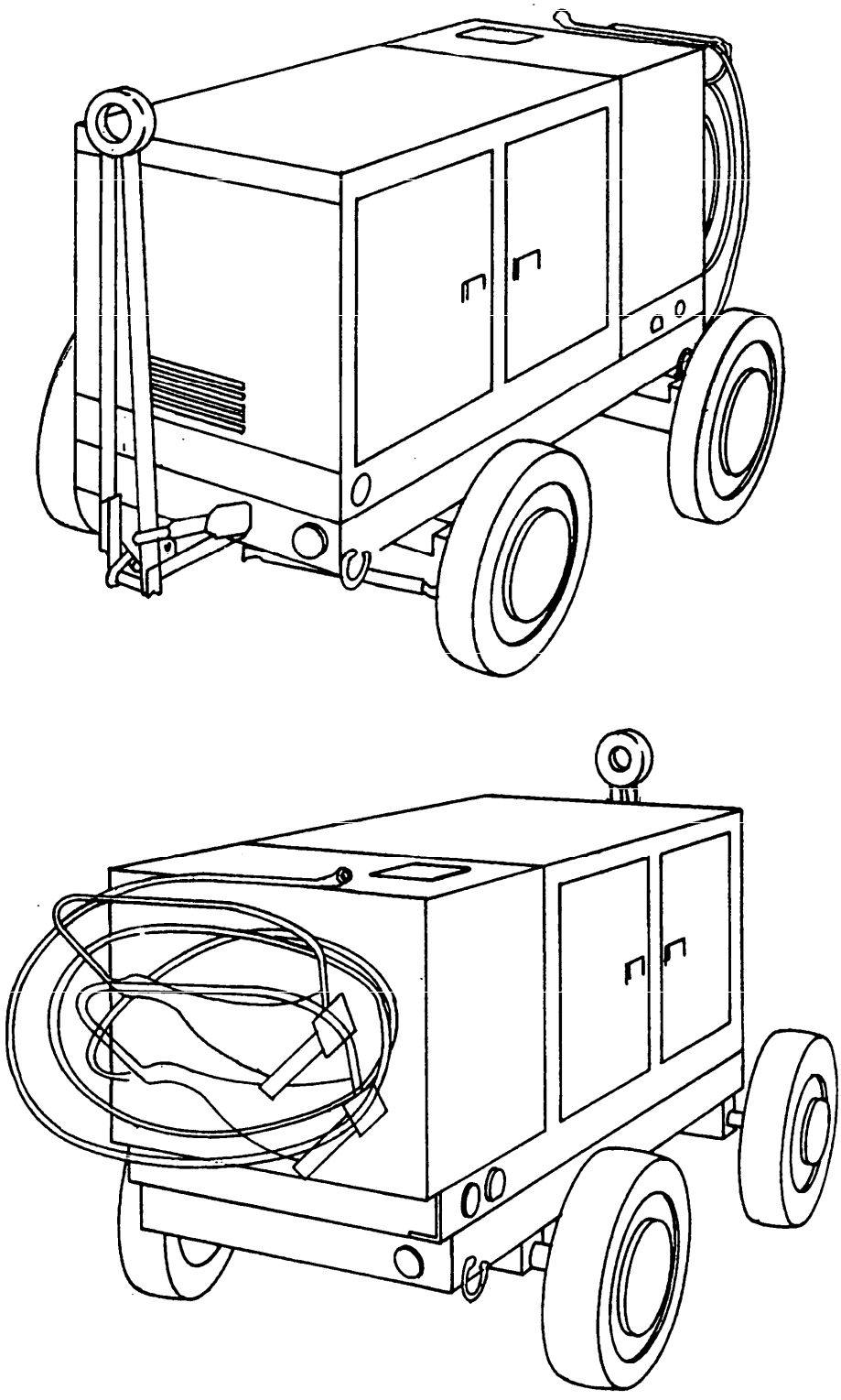


Figure 1. Cleaning Machine Outline Drawing (Representative)

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3.5.4 Screw threads - Screw threads shall be in accordance with MIL-S-7742.

3.5.5 Castings and forgings - All castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, mismatching and other defects that may affect the structural strength of the forgings involved. Strength and other essential physical and chemical properties of the castings and forgings shall be adequate to meet the performance requirements specified herein. In no event shall welding, brazing, soldering, peening or plugs, be used on castings or forgings for reclaiming any parts of the cleaning machine without authorization of the procuring activity.

3.5.6 Static electricity - The machine shall be equipped with suitable grounding devices for static discharge.

3.5.7 Fastening devices - All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently deformed.

3.5.8 Grease fittings - Unless otherwise specified, grease fittings conforming to MIL-F-3541 shall be provided in accessible positions at all necessary lubrication points. Pressure relief fittings shall be provided where use of high pressure lubrication equipment could cause damage.

3.5.9 Safety devices - The cleaning machine shall be equipped with safety devices to adequately protect the operator from all hazards incident to its operation.

3.5.10 Maintainability - The cleaning machine shall conform to the ease of operation, maintenance, and repair requirements specified in MIL-M-8090.

3.5.11 Air transportability - The cleaning machine shall be designed for air transportability in accordance with MIL-A-8421.

3.6 Performance characteristics - The cleaning machine and component parts shall meet the performance tests in Section 4 in the order listed below:

Trailer tests	(4.7.2)
Pump protection test	(4.7.3)
Water source test	(4.7.4)
Water temperature test	(4.7.5)
Cleaning compound tank leak test	(4.7.6)
Cleaning compound-water ratio test	(4.7.7)
Variable water pressure test	(4.7.8)
Spray gun nozzle test	(4.7.9)

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Spray gun shock test	(4.7.10)
Endurance test	(4.7.11)
High pressure hose test	(4.7.12)
Foam application test	(4.7.13)
Cleaning compound transfer test	(4.7.14)
Enclosure strength test	(4.7.15)
Forklift test	(4.7.16)
Hoisting test	(4.7.17)
Electromagnetic interference test	(4.7.18)

3.7 Details of components - The mobile aircraft cleaning machine shall include, but not be limited to, the following components:

Trailer	(3.7.1)
Engine	(3.7.2)
Clutches	(3.7.3)
Intake hoses	(3.7.4)
Hose adapter	(3.7.5)
Pressure cleaning system	(3.7.6)
Foam system	(3.7.7)
Cleaning compound transfer equipment	(3.7.8)

3.7.1 Trailer - The trailer shall conform to the requirements of MIL-M-8090, type I, group C, class 2. It shall completely enclose all component parts except the hose rack and hose, which shall be mounted outside the enclosure. Provisions shall be made to insure that water does not collect on any surfaces of the trailer. It shall be equipped with a drawbar. The center of gravity, with all components installed, shall be a maximum of 24 inches from the ground. Refer to Figure 1 for a representative illustration.

3.7.1.1 Ground clearance - Trailer ground clearance shall be as specified in MIL-M-8090, type 1, class 2.

3.7.1.2 Control panel - All gages, switches, meters, and controls shall be mounted on a single control panel. The panel shall be located inside the enclosure.

3.7.1.3 Doors - The enclosure shall include hinged doors to provide easy access to the control panel. The doors shall be capable of being secured in the closed position and a positive means shall be provided to hold the doors in the open position.

3.7.1.4 Wheels and hubs - The wheels and hubs shall be as specified in MIL-W-8005.

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3.7.1.5 Tires - The tires shall be as specified in MIL-T-83926, type VII, class NDMS, size 480/400 - 8, 4 ply.

3.7.1.6 Parking brakes - The rear wheels shall be equipped with parking brakes conforming to the requirement of MIL-M-8090. The brakes shall be actuated by a manual control lever located on the front end of the trailer adjacent to the towbar. The brakes shall be capable of being locked and released by use of the brake control lever.

3.7.1.7 Towing provisions - Towing provisions shall be designed in accordance with MIL-M-8090. The towbar shall be three feet long, capable of supporting a 500 pound load placed at any point when it is in a horizontal position. The towbar shall be equipped with a lunette ring conforming to MS51336. A latch shall be provided for stowing the drawbar in a vertical position when not in use. The latch shall be so designed that the drawbar cannot be removed from the vertical position without manually actuating the latch.

3.7.1.8 Light reflectors - Light reflectors, conforming to the requirements of MIL-R-12144, shall be installed on the trailer as specified in MIL-M-8090.

3.7.1.9 Hose racks - The trailer shall be equipped with two hose racks (one for the high pressure hose and one for the foam hose). Each rack shall be capable of housing 100 feet of hose.

3.7.1.10 Spray gun storage - A safe and convenient place shall be provided on the trailer to secure the spray guns during transit and when not in use.

3.7.1.11 Storage area - A storage area shall be provided on the trailer. The area shall be protected from the weather and capable of securely housing the hose adapter (3.7.5), cleaning compound transfer equipment (see 3.7.8) and the necessary tools and accessories.

3.7.1.12 Hoisting and tiedown - The trailer shall be equipped with not less than four tiedowns as specified on Air Force Drawing 48B7796. The tiedown devices shall serve for both tiedown and hoisting, and shall be located around the periphery of the unit in a horizontal plane. The rings shall be so located that Safe Slings can be rigged from a common cable and spreader bar for hoisting by a single-hook overhead crane.

3.7.2 Engine - The cleaning machine shall be equipped with a gasoline engine conforming to the requirements of MIL-E-11275, group I, four cycle. The engine shall be equipped with a 12 volt ignition system and an electric cranking system.

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The engine shall also be equipped with the following components:

(a) Twelve volt battery in accordance with MS35000

(b) Voltage regulator

(c) Generator

(d) Time totalizing meter in accordance with MIL-M-7793

(e) Fuel tank (minimum capacity 5 gal)

3.7.2.1 Drain plug - The fuel tank shall be equipped with a drain plug capable of draining the fuel from the tank (necessary for air transport).

3.7.3 Clutch - The cleaner shall be equipped with an electro-magnetic clutch having a minimum static torque rating of 600 inch pounds.

The clutch shall transfer power from the engine to the clear water pump. It shall also permit the engine to be started in an unloaded condition.

3.7.4 Intake hoses -

3.7.4.1 Intake hose (low pressure - 3/4" ID) - The cleaning machine shall be equipped with 50 feet of 3/4 inch diameter (ID) hose conforming to the requirements of ZZ-H-611. The hose shall be used to conduct water from a pressure source (hydrant) to the input side of the water pump. The hose shall be equipped with a coupling conforming to the requirement of WW-C-624, type A.

3.7.5 Pressure cleaning system - The pressure cleaning system shall include but not be limited to the following components:

Cleaning compound tank	(3.7.6.1)
Clear water - cleaning compound ratio control	(3.7.6.2)
Water pump	(3.7.6.3)
Check valves	(3.7.6.5)
Toggle valves	(3.7.6.6)
High pressure hose	(3.7.6.7)

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Hose assemblies	(3.7.6.8)
Spray gun	(3.7.6.9)

3.7.5.1 Cleaning compound tank -

3.7.5.1.1 Material - The steel used in the construction of the cleaning compound tank shall conform to the requirement of QQ-S-766, class 304, 16 gage minimum thickness, finish 2D or 2B.

3.7.5.1.2 Capacity - The cleaning compound tank shall have a capacity of 58 gallons plus a minimum of 4 inches of freeboard. The tank shall be equipped with a gage calibrated in five gallon increments. When empty, the tank shall not "oil can".

3.7.5.1.3 Tank cover - The cleaning compound tank shall be equipped with a liquid-tight, hinged cover. It shall permit access to the tank interior for normal maintenance and cleaning purposes and shall be self-supporting in the open position.

3.7.5.2 Clear water-cleaning compound ratio control - The high pressure cleaning system shall be equipped with a ratio control mechanism capable of delivering a mixture of clear water and cleaning compound in the following proportions.

Cleaning compound	Clear water
1	5
1	10
1	20

3.7.5.2.1 Control mechanism - The ratio control mechanism shall be compatible with compounds conforming to the requirements of TT-C-4901, TT-P-664, VV-K-211, MIL-C-43616, and MIL-C-38334.

3.7.5.3 Water pump - The water pump shall be capable of pumping water at 10 gpm ± 1/2 gpm at 500 psig. The pump shall be equipped with a pressure regulator, pressure gage and a by-pass valve. The bypass line shall terminate at the input side of the pump. The pump shall be impervious to cleaning compound conforming to the requirements of MIL-C-43616.

3.7.5.3.1 Pressure regulator - The output side of the pump shall be capable of being varied by a simple mechanical adjustment.

3.7.5.3.2 Drain cock - The pump shall be equipped with a drain cock, located at the lowest point on the pump. It shall be capable of being operated without the use of tools.

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3.7.5.4 In addition to the drain cocks on the water pump, the lowest point in the system shall also be equipped with a drain cock. The drain cocks are for use in draining the system prior to air transport and to preclude damage to the system when it is exposed to temperatures below 32° F.

3.7.5.5 Check valves - The pressure cleaning system shall be equipped with check valves strategically located in the system to prevent clear water from being pumped into the cleaning compound tank. The system shall also be equipped with a check valve to prevent a reverse flow of water through the pump.

3.7.5.6 Toggle valves - The pressure cleaning system shall be equipped with two manually operated toggle valves (no electrical components). The valves shall be mounted on the trailer and function as shutoff valves (interrupting the flow of cleaning compound from the tank and clear water from the pump).

3.7.5.7 High pressure hose - The pressure cleaning system shall be equipped with 100 feet of high pressure hose conforming to the requirements of MIL-H-8794.

3.7.5.8 Hose assemblies - Hose assemblies shall be in accordance with MIL-H-8795.

3.7.5.9 Spray gun - The spray gun shall have a pistol grip handle (Spraying Systems Co., Bellwood, Ill. 60104, model 43HC 1/2-unijet nozzle and a #11990-13 swivel, or equal). The extension length shall be 36 inches.

3.7.5.9.1 Controls - The spray gun shall be capable of producing a spray pattern ranging from a solid stream to a spray angle (included angle) of 40°. The controls shall also insure a drip-free shut-off.

3.7.5.9.2 Neoprene protectors - The spray gun and extension shall be equipped with neoprene spheres positioned on the gun (and extension) to protect it from damage when the gun is dropped or dragged on a concrete surface.

3.7.6 Foam system - The foam system shall be capable of applying the cleaning compound as a foam to the external surfaces of aircraft.

3.7.6.1 Components - The foam system shall include but not be limited to the following components:

Spray foam unit	(3.7.6.2)
Valves	(3.7.6.3)
Air inlet fitting	(3.7.6.4)
Hoses	(3.7.6.5)
Spray gun	(3.7.6.6)

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3.7.6.2 Spray foam unit - The spray foam unit shall be a Dema Engineering Company, St. Louis, Mo. 63122 spray foam model no. 293, or equal. It shall be installed in an accessible position on the trailer to facilitate adjustment of the controls.

3.7.6.3 Valves - The system (piping) shall include 3 manually operated shut-off valves; one for clear water, one for cleaning compound and one for compressed air.

3.7.6.4 Air inlet fitting - The air inlet port of the foam unit shall be equipped with a quick-disconnect fitting (Hansen Mfg. Co., Cleveland, Ohio, series 3,000; or equal).

3.7.6.5 Hoses - The foam unit shall be equipped with a 100 feet of hose (minimum diameter 1") Gates Rubber Co., Denver, Colorado, 80209; or equal.

3.7.6.6 Spray gun - A spray gun shall be attached to the hose specified in 3.7.6.5. It shall be equipped with a shut-off valve, two nozzles (solid stream and vee-jet). The extension on the gun shall be 48 inches long.

3.7.6.7 Swivel connection - The connection between the spray gun and the hose shall permit the gun to swivel on the hose.

3.7.7 Transferring cleaning compound from a 55 gallon drum to cleaning compound tank (capability) - To expedite the transfer of cleaning compound from a 55 gallon drum to the cleaning compound tank the cleaning machine shall be equipped with fifteen feet of hose conforming to the requirements of ZZ-H-611. The hose shall be equipped with a quick-disconnect type fitting for attaching the hose to the input side of the pump. A metal tube equipped with a strainer and a foot check valve shall be provided. The strainer and the metal tube shall be equipped with quick-disconnect type fittings for attaching to the hose. The metal tube shall fit through the two inch diameter opening in the drum and be long enough to completely empty the drum.

The cleaning machine shall be equipped with permanent piping to conduct the cleaning compound from the output side of the pump to the cleaning compound tank. The necessary controls (manual) shall be incorporated into the piping to prevent clear water (during normal operation) from entering the cleaning compound tank and also to preclude pumping the cleaning compound directly into the high pressure hose or into the foam system. The cleaning compound shall be delivered into the storage tank in a completely liquid state (absence of foam).

3.8 Finish - All metal surfaces of the cleaning machine on which a protective coating will not be detrimental to the operation of the machine shall be painted.

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3.8.1 Exterior surfaces -

(a) Pretreatment in accordance with TT-C-490, Type 1.

(b) One coat of primer in accordance with TT-P-664.

(c) Two coats of gloss enamel in accordance with MIL-E-7729. Color shall be in accordance with FED-STD-595, Color No. 13538.

3.8.2 Interior surfaces and equipment - The interior surfaces of the cleaning machine, including engine, pump, and other components shall be painted as specified in MIL-M-8090.

3.9 Part numbering of interchangeable parts - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.10 Identification of product -

3.10.1 Identification plate - A corrosion-resistant metal nameplate shall be permanently attached to the cleaning machine. The identification marking shall be in accordance with MIL-STD-130.

3.11 Control marking, operating instructions, transportation data plate - Control marking, operating instructions, and the transportation data plate shall be as specified in MIL-M-8090.

3.12 Workmanship - The general appearance shall reflect the highest standard of workmanship, uniformity of quality, and freedom from defects or foreign matter which could adversely affect safety, performance, reliability, or durability of the cleaning machine.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Inspection terms - MIL-STD-109 shall govern the definition of terms used herein.

4.2 Classification of inspections - The inspection of the cleaning unit shall be classified as follows:

(a) First article inspection (4.3)

(b) Quality conformance inspection (4.4)

4.3 First article inspection - The first article inspection shall consist of all the inspections of this specification.

4.3.1 First article sample - Unless otherwise specified, as soon as practicable after the award of the contract or order, the manufacturer shall submit one unit for inspection to determine conformance with this specification (see 6.2.1.1). The unit shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these units from contract to contract, submission of a first article sample on the new contract may be waived at the discretion of the Government (see 6.2.1.1). Approval of the first article sample or the waiving of the first article inspection is not a waiver of the requirement of submitting to the quality conformance inspection.

4.3.1.1 Upon completion of the first article inspection, all the applicable inspection reports, and when applicable, recommendations and comments pertinent for use in monitoring production shall be forwarded to the cognizant government activity (see 6.2). The approved unit shall be returned to the manufacturer for use in monitoring production.

4.4 Quality conformance - Each cleaning machine on the contract or order shall be subjected to the following inspections in the order listed.

Examination	(4.7.1)
Braking test	(4.7.2b)
Cleaning compound-water ratio test	(4.7.8)
Endurance test (except that time period (step h) shall be 15 minutes)	(4.7.12)

4.5 Test conditions - Unless otherwise specified, tests shall be conducted at an ambient temperature of 77 ± 18° F and at an ambient barometric

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pressure of 28 to 32 inches of mercury. If tests are conducted at different ambient conditions, appropriate correction factors shall be applied to the flow and pressure readings.

4.5.1 Inspection sampling - The quality conformance examinations and tests shall be performed on each cleaning machine offered for acceptance under the contract or order.

4.6 Inspection of materials -

4.6.1 Commercial materials - When materials are procured under commercially recognized non-governmental specifications, such as SAE numbers, a certificate showing compliance with such specifications shall be furnished by the supplier of the material. Such a certificate shall be in a form acceptable to the Government inspector.

4.7 Inspection methods -

4.7.1 Examination - The cleaning unit shall be examined to determine conformance with the requirements in Section 3 that are not covered by specific tests.

4.7.1.1 The cleaning machine shall be examined to ascertain that the preparation for delivery conforms to this specification.

4.7.2 Trailer tests - With the cleaning machine mounted on the trailer, cleaning compound tank filled to design capacity with water, and the tank cover in the closed position, the trailer shall be subjected to, and pass, the following tests specified in MIL-M-8090.

(a) Servicing and maintenance test

(b) Braking test

(c) Ramp test

(d) Mobility test (Type I, Class 2)

There shall be no evidence of leakage from the tank cover during the mobility test.

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4.7.3 Pump protection test - The pump protection test shall be conducted as follows:

(a) Place the cleaning machine on a plane of 0° with the horizontal.

(b) Attach the intake hose to a water source having a minimum pressure of 30 psig and a minimum flow of 10 gallons per minute (gpm).

(c) Place the water intake valve in the open position.

(d) Operate the pressure cleaning system for two minutes.

(e) Close the water intake valve.

The pump shall stop automatically and there shall be no damage to the pump.

4.7.3.1 Continue the pump protection test as follows:

(a) Place the cleaning machine on a plane 15° with the horizontal.

(b) Repeat steps (c), (d), and (e) of para. 4.7.3.

The pump shall stop automatically and there shall be no damage to the pump.

4.7.4 Water source test - The water source test shall be conducted as follows:

(a) Repeat steps (a), (b), (c) of para. 4.7.3.

(b) Operate the pressure cleaning system for one hour.

There shall be no evidence of damage to, or malfunction of, the cleaning machine during the test.

4.7.4.1 Continue the water source test as follows:

(a) Place the cleaning machine on a plane 15° with the horizontal.

(b) Repeat steps (b), (c) of para. 4.7.3.

(c) Operate the pressure cleaning system for one hour.

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There shall be no evidence of damage to, or malfunction of, the cleaning machine during the test.

4.7.5 Water temperature test - The water temperature test is performed with the cleaning machine on a plane of 0° with the horizontal, and on a plane of 15° with the horizontal. In each of the two positions the test shall be conducted as follows:

(a) Attach the water intake hose (3/4" ID) to a water source having a temperature of 100° F ± 10° F, a pressure of 30 psig, and a minimum flow rate of ten gallons per minute (gpm).

(b) Operate the cleaning machine for a period of one hour.

There shall be no damage to, or malfunction of, the cleaning machine during the test.

4.7.6 Cleaning compound tank leak test - With the cleaning machine on a plane of 0° with the horizontal, fill the cleaning compound tank to design capacity with water, and permit the water to remain in the tank for one hour. There shall be no evidence of leakage from the tank.

4.7.7 Cleaning compound-water ratio test - The cleaning compound-water ratio test shall be performed with the cleaning machine on a plane of 0° with the horizontal. The test shall be conducted as follows:

(a) Attach the water intake hose (3/4" ID) to a water source having a minimum pressure of 30 psi, and a minimum flow of 10 gpm.

(b) Place ten gallons of kerosene in the cleaning compound tank.

(c) Set metering device to achieve a ratio of one part kerosene to five parts water.

(d) Apply power to the pump.

(e) Using a transparent beaker, collect 1000 ml of the effluent from the nozzle of the hose wand.

(f) Add 1/8 fluid ounce of propylene glycol, color red (McCormick and Co., Baltimore, Md., food coloring, or equivalent), into the beaker and stir until all the fluid is colored red.

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(g) Permit the fluid in the beaker to remain motionless until there is a definite interface of red and clear fluids (approximately fifteen minutes).

(h) Repeat steps d, e, f and g with the kerosene - water ratios as follows:

Kerosene	Water
1	10
1	20

(i) The ratio of kerosene to water is determined by the volume of clear fluid (kerosene) to the volume of red colored fluid (water).

(j) The cleaning compound - water ratio, as measured, shall be ±10% of the indicated meter settings.

4.7.8 Variable water pressure test - The variable water pressure test shall be performed on a plane of 0° with the horizontal. The test shall be conducted as follows:

(a) Set the cleaning compound metering device to achieve a ratio of one part kerosene to ten parts water.

(b) Within a time frame of one hour, vary the pressure (a minimum of 10 times) on the output side of the pump between 400 and 200 psi.

The ratio of kerosene to water shall be as specified in 4.7.8, step (j).

4.7.9 Spray gun nozzle test - Place the nozzle in the full open position. The gun shall discharge a solid stream measuring approximately 1-1/2 inches in diameter at a distance of 36 inches from the nozzle.

4.7.9.1 Place the nozzle in the full feathered position. It shall discharge a fan shaped stream of water measuring approximately 12 inches wide at a distance of 6 inches from the nozzle.

4.7.10 Spray gun shock test - With the spray gun attached to the high pressure hose, drop the gun from a height of 20 feet to a hard concrete surface. Repeat 5 times.

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There shall be no evidence of deformation or damage to the gun; in addition, the gun shall be subjected to and pass the tests specified in 4.7.9 and 4.7.9.1.

4.7.11 Endurance test - With the cleaning compound tank filled to design capacity with cleaning compound conforming to the requirements of MIL-C-43616, operate the cleaning machine as follows:

(a) Connect the water intake hose to a water source having a pressure of 40 psi ±10 psi (standard house faucet and water pressure).

(b) Open the faucet valve.

(c) Position the controls to establish a water-to-cleaning compound ratio of 5 to 1.

(d) Place the water intake valve in the open position.

(e) Place the spray gun valve in the open position.

(f) Start the engine.

(g) Achieve a pressure of 350 psi on the output side of the pump.

(h) Operate the cleaning machine for a period of two hours.

(i) Close the spray gun valve.

(j) Stop the engine.

During the entire test there shall be no evidence of malfunction or failure.

4.7.12 High pressure hose test - The high pressure hoses shall be tested as specified in FED-STD-601, test method 10211. The hoses shall have a minimum burst pressure of 2000 psi.

4.7.13 Foam application test - The foam application test shall be conducted as follows:

(a) Position a steel or aluminum plate (12 feet square) in a vertical plane three feet above ground level.

(b) Connect the foam unit to an external (minimum pressure 100 PSI) air source.

(c) Adjust the controls on the foam unit.

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(d) Apply foam to the (12 feet square) plate.

The foam shall completely cover; and cling to the plate for a period of time not less than one minute.

4.7.14 Cleaning compound transfer test (from 55 gallon drum to cleaning compound tank) - The transfer test shall be conducted as follows:

(a) Connect the cleaning compound transfer equipment (specified in 3.7.7) to a 55 gallon drum of cleaning compound (conforming to the requirements of MIL-C-43616) and the input side of the pump.

(b) Position piping controls (valves) so that the cleaning compound will be transferred directly to the cleaning compound tank.

(c) Apply power to the pump.

The contents of the drum shall be transferred to the tank within a time period not to exceed 5 minutes.

4.7.15 Enclosure strength test - Weights equivalent to 250 pounds per square foot shall be placed at various locations on the top of the enclosure for a period of one hour. There shall be no evidence of structural deformation or damage to the enclosure.

4.7.16 Forklift test - Using a forklift truck conforming to the requirements of MIL-T-21863, the cleaning machine shall be lifted to a height not less than three feet from the ground and transported a distance of ten yards and placed on the ground. This procedure shall be repeated five times. There shall be no evidence of structural deformation or damage to the cleaning machine.

4.7.17 Hoisting test - The hoisting test shall be conducted using an overhead crane and the hoisting (tiedown) rings. The cleaning machine shall be lifted a minimum of three feet from the ground, suspended at this height for a minimum of three minutes, and lowered to the ground. This procedure shall be repeated five times. There shall be no evidence of deformation or damage to the cleaning machine.

4.7.18 Electromagnetic interference test - The electromagnetic interference test shall be conducted in accordance with MIL-STD-461, Class IIIc.

5. PREPARATION FOR DELIVERY

5.1 Cleaning, preservation and packaging - Cleaning and preservation, excluding the engine, fuel tank and battery shall be in accordance with the requirements of MIL-P-116 (see 6.2.b). No packaging is required.

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5.1.1 Engine, fuel tank and battery -

5.1.1.1 Overseas shipment - For overseas shipment, the engine, fuel tank. and battery shall be preserved in accordance with MIL-E-10062, Type II, Level A, Method 1.

5.1.1.2 Domestic shipment - For domestic shipment, the engine, fuel tank, and battery shall be preserved in accordance with MIL-E-10062, Type III, Level C.

5.1.3 Water pump - The interior surfaces of the water pump shall be flushed with a commercial type permanent type anti-freeze solution (60% anti-freeze - 40% water). After flushing, the pump shall be gravity drained.

5.1.4 Electrical connections - Exposed electrical connections shall be coated with varnish conforming to MIL-V-173.

5.2 Packing - Packing shall be level A, B, or C as required.

5.2.1 Level A - After being cleaned and preserved as specified in 5.1, the cleaning machine shall be packed in containers conforming to the requirements of PPP-B-601 or PPP-B-621 (overseas type).

5.2.1.1 Blocking and bracing - Blocking, bracing and cushioning shall be in accordance with MIL-STD-1186.

5.2.2 Level B - The machine, after being cleaned and preserved as specified in 5.1, shall be packed as indicated in 5.2.1 except that the container shall be of the domestic type.

5.2.3 Level C - After being cleaned and preserved as specified in 5.1, the cleaning machine shall be packed to insure carrier acceptance and safe delivery to destination. Packing shall comply with Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.2.4 Hoses -

5.2.4.1 Intake hose - The intake hose (3.7.4.1) shall be compactly coiled and secured to the cleaning machine with tape conforming to PPP-T-76.

5.2.4.2 High pressure hose - The high pressure hose (3.7.5.7), with spray gun connected, shall be compactly coiled and secured to the cleaning machine with tape conforming to PPP-T-76.

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5.2.4.3 Foam hose - The foam hose (3.7.6.5) with spray gun connected shall be compactly coiled and secured to the cleaning machine in the tape conforming to PPP-T-76.

5.2.5 Towbar - The towbar shall be disconnected from the cleaning machine and secured to the bottom of the container.

5.3 Marking - In addition to any special marking required in the contract or order (see 6.2), marking of packages and containers shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use - The cleaning machine covered by this specification is intended for use on naval shore installations and airfields to remove salt deposits, oil, grease and other contaminants from carrier type aircraft and helicopters.

6.2 Information for the contracting officer - Contracts or orders should specify the following:

(a) Title, number, and date of this specification

(b) Selection of applicable level of preservation, packaging, and packing required (see 5.1, 5.2, and 5.3)

(c) Special marking, if required (see 5.3)

(d) Items of data required (see 6.3)

(e) First article inspection, if other than specified (6.2.1.1)

(f) Where first article inspection shall be conducted (6.2.1.1.a)

(g) Where first article inspection report shall be sent

(h) Engineering change procedure shall be in accordance with NAVAIR INST 4275.3A. The Naval Air Systems Command, Dept. of the Navy, Washington, D.C. 20360 (attn: AIR-534), will be responsible for approval of class I changes under the contract. The Commanding Officer, Naval Air Engineering Center, Philadelphia, Pa. 19112 (attn: GSED) will be responsible for approval of class II changes.

6.2.1 Contracts or orders should specify the following provisions for first article inspection.

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6.2.1.1 Whether first article inspection is required - When a contractor is in continuous production of the cleaning machine from contract to contract, consideration should be given to waive the first article inspections. If first article inspection is required, indicate:

(a) Where the first article inspection is to be conducted (at the contractor's plant or Government or commercial laboratory).

(b) That the approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.3 Data - For the information of Contractors and Contracting Officers, any of the data specified in subparagraphs below, applicable documents listed in Section 2 of this specification, or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NAVAIR Form 4200/25 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

6.3.2 Manuals - Operation and maintenance manuals shall be written in accordance with MIL-M-38784.

6.4 Supersession data - This specification includes the requirements of purchase description XWS-5527, dated 15 September 1965.

Preparing Activity
Navy - AS
(Project No. 4940-N237)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-C-81825(AS) CLEANING MACHINE, SMALL AIRCRAFT, MOBILE

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426

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Naval Air Systems Command
Washington, D.C. 20360

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